# Lectures

## Stat 154/254: Statistical Machine Learning

**Lectures marked with $\star$ are still in progress.**



## Introduction

- Class introduction
- Key concepts

## Unit 1: Regression in a machine learning setting

- Linear regression in matrix notation (review)
- Linear regression in an ML setting
- Basis expansions and splines for linear regression
- L1 and L2 penalization

## Unit 2: Classification

- Empirical risk minimization for classification
  - Perceptron algorithm **(extra, not covered in class)**
- Trading off false positives and negatives

## Unit 3: Risk and complexity

- Regret decomposition
- Asymptotics and bias–variance tradeoff in linear regression
- Generalization bounds
  - Generalization error and uniform laws for smooth function classes
  - VC dimension and zero–one loss
- Cross validation
  - ⋆ Held-out data and cross-validation
  - ⋆ Simulated bias–variance tradeoff example
  - ⋆ Cross–validation cautionary note

## Unit 4: Trees and weak learners

- ⋆ Classification and regression trees
- ⋆ Bagging weak learners
- ⋆ Boosting weak learners

## Unit 5: Neural networks and optimization

- ⋆ Optimization notes

### Prerequisites

Mathematical statistics, probability, linear algebra, multivariate calculus, and Python are all prerequisites.

For assistance with mathematics prerequisites, the Student Learning Center provides additional assistance.

A summary of the required linear algebra skills can be found here.